

Design optimization of the insulation domain in a building wall

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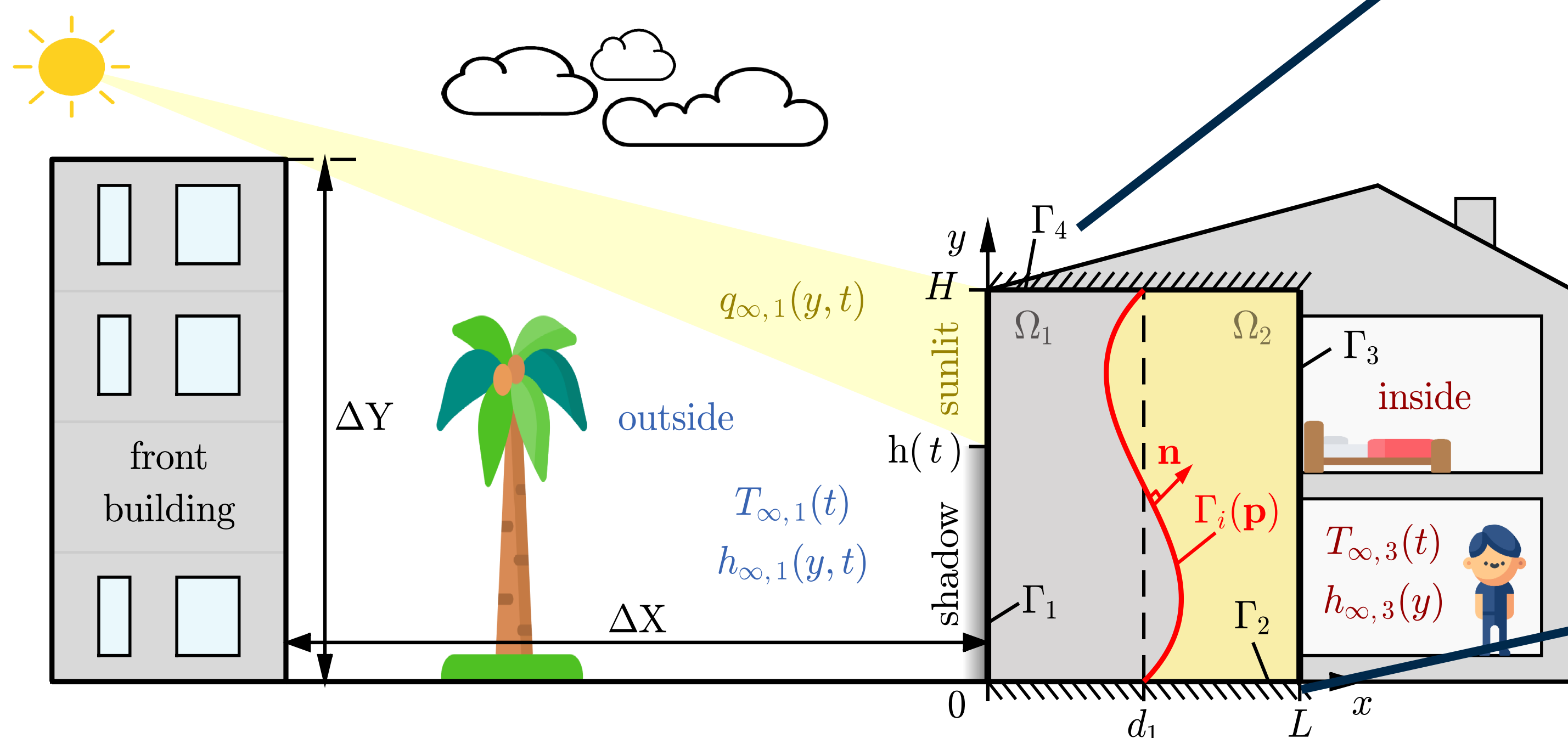
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Introduction

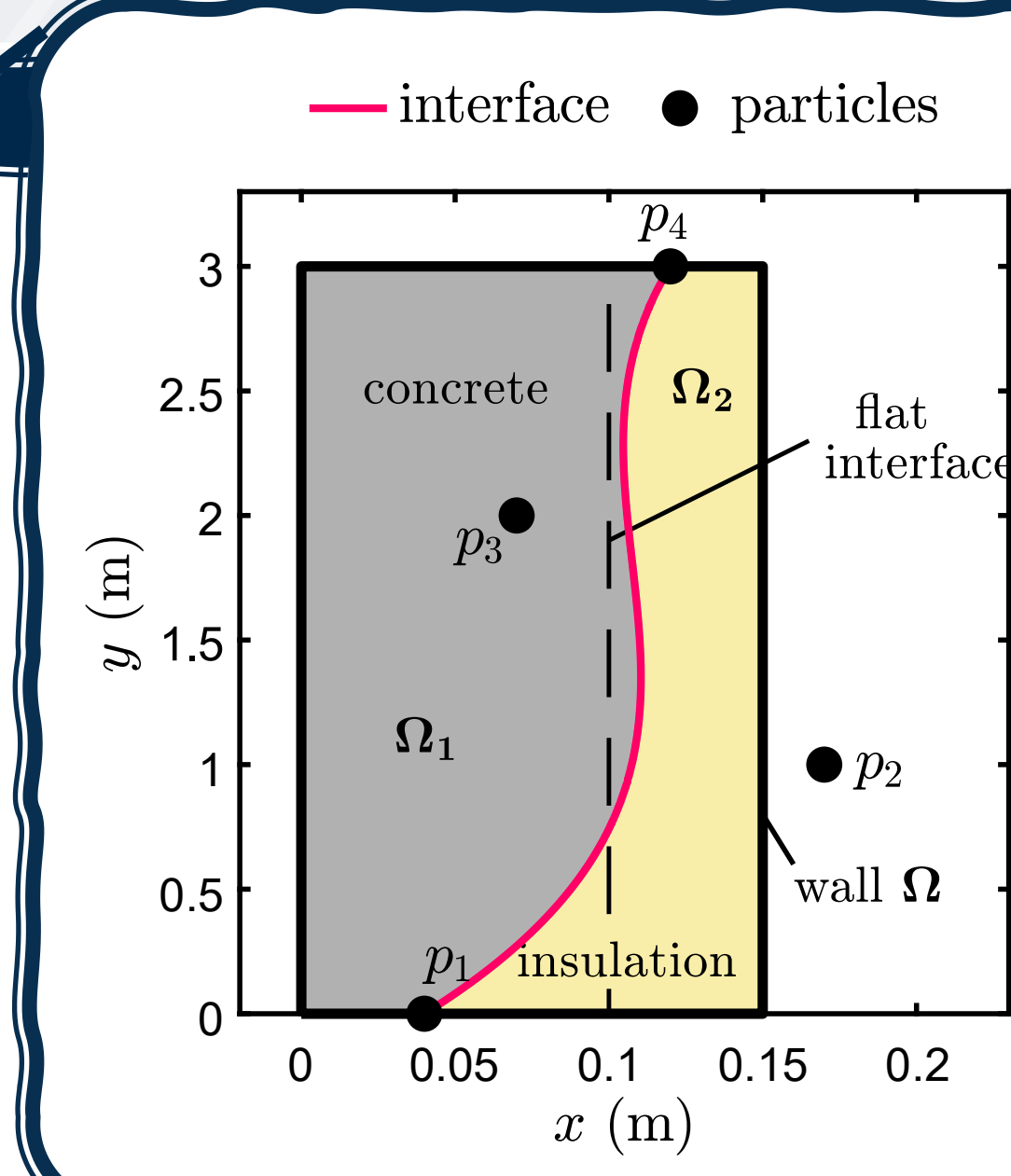
Context: Building envelopes account for a significant share of global energy consumption, making energy efficiency a key challenge. Conventional wall design approaches are typically based on simplified assumptions [1], such as uniform solar radiation and planar material interfaces, which overlook important spatio-temporal heterogeneities, including wind variability, shading effects, and time-dependent boundary conditions. Recent advances, such as additive manufacturing in construction, now enable complex geometries [2]. However, most optimization studies remain limited to steady-state analyses [3], neglecting the inherently transient nature of real environmental loading conditions.

Objectif: This work presents a novel optimization framework for thermal wall design by geometrically optimizing the interface between structural and insulating materials. The objective is to minimize energy demand.

Physical model



- Two-dimensional building wall located in an urban canyon environment.
- Transient heat transfer including convection and shortwave radiation on Γ_1 , convection on Γ_3 , and adiabatic boundary conditions on Γ_2 and Γ_4 . At the interface Γ_i between the two materials, continuity of conductive heat flux and equality of temperature are enforced.



Interface definition

The interface Γ_i is represented using a Bézier curve defined with 4 control points and Bernstein polynomial interpolation:

$$f_{B_x}(s, \mathbf{p}) = \sum_{i=1}^N B_{i-1}^{N-1}(s) \cdot x_i + A(\mathbf{p})$$
$$f_{B_y}(s, \mathbf{p}) = \sum_{i=1}^N B_{i-1}^{N-1}(s) \cdot y_i, \quad \forall s \in [0, 1]$$

Thermal design problem

The minimization problem aims to find the optimal coordinates of the control points $\mathbf{p}^* = \min_{\mathbf{p} \in \Omega_p} \mathcal{C}$ of the Bézier curve, defined as:

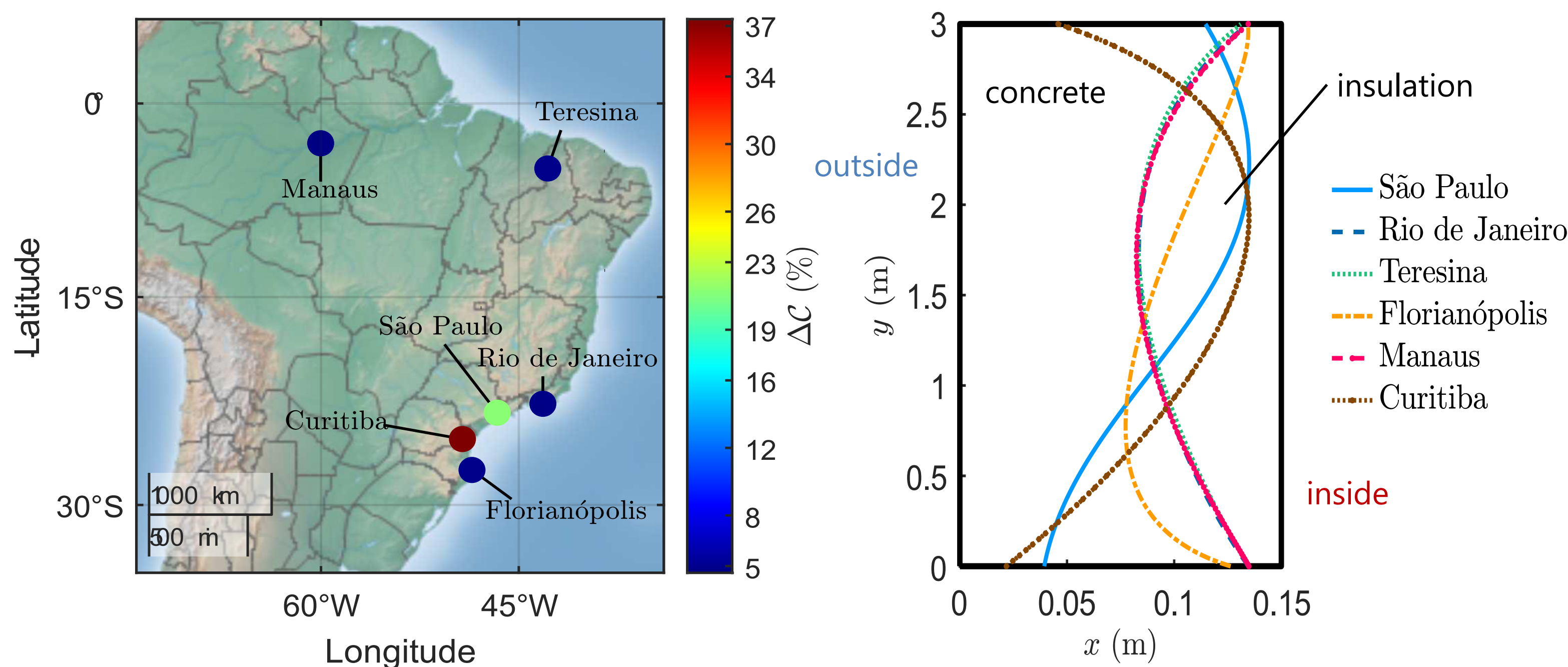
$$\mathcal{C} = \int_0^{t_f} w(t) dt, \quad \text{with } w(t) = \left(\int_{\Gamma_3} -\lambda(\mathbf{x}) \cdot \frac{\partial T}{\partial x} dy \right) \cdot \left(\frac{T_{\infty,1}(t)}{T_{\infty,3}(t)} - 1 \right)$$

subject to:

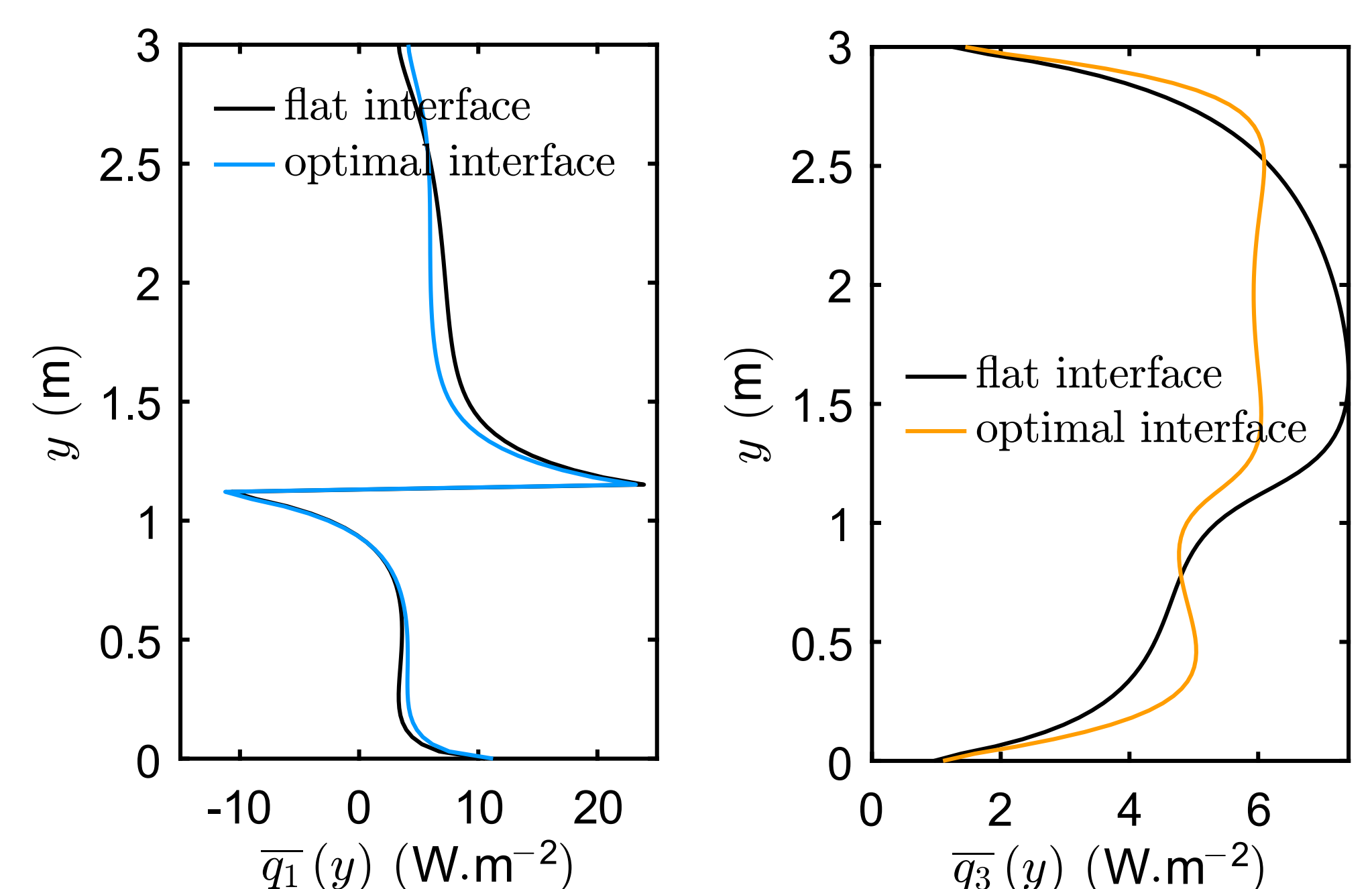
- Volume constraint: layer volumes remain constant.
- Spatial constraint: the interface must lie within the wall boundaries;

Convergence of the NSGA-II optimization is assessed using $\delta\mathcal{C}$ (relative objective variation) and δd (mean Euclidean parameter dispersion), with: $\varepsilon_g = \max(\delta\mathcal{C}_g, \delta d_g) \quad \forall g \in \{2, \dots, N_g\}$

Case study in Brazil



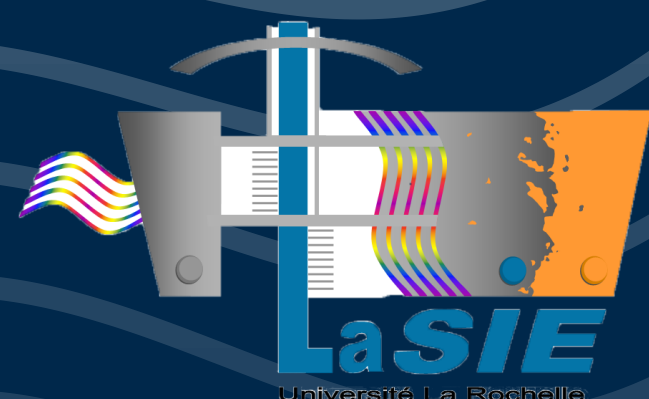
- Promising performance in the few Brazilian cities tested under conditions of low internal insulation.
- The optimal interface in Rio de Janeiro redistributes and reduces the average internal heat flux $\bar{q}_3$ across the entire height of the wall, and reduces the magnitude of the average external heat flux $\bar{q}_1$ compared to a flat interface.



Conclusion & perspectives

- Annual energy improvements of up to nearly 40% are achieved for the north-oriented façade compared with conventional flat interfaces, highlighting the potential of interface geometry optimization.
- The study is limited to Bézier curves with N=4 control points, restricting geometric complexity. Building-level thermal response is not simulated, and coupling with dynamic thermal simulation software is needed for more accurate predictions.

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References

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